EKACTL-Typst, section 6

Linear algebra

Let $\mathbb{F}$ be a field. It's possible that some things in here require $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$; if it does and it's not marked it should be considered a bug.

1. Fields

Definition 1.1 (Field properties)

The following properties:

Theorem 1.2 (Properties of addition)

For all $x, y, z \in \mathbb{F}$:
- (A1) Commutativity: $x + y = y + x$
- (A2) Associativity: $(x + y) + z = x + (y + z)$
- (A3) Identity: $\exists 0 \in \mathbb{R}$ s.t. $0 + x = x$
- (A4) Additive inverse: For $x \in \mathbb{F}$, $\exists -x \in \mathbb{F}$ s.t. $x + (-x) = 0$

Theorem 1.3 (Properties of multiplication)

For all $x, y, z \in \mathbb{F}$:
- (M1) Commutativity: $xy = yx$
- (M2) Associativity: $(xy)z = x(yz)$
- (M3) Identity: $\exists 1 \in \mathbb{R}$ s.t. $1x = x$ and $1 \neq 0$
- (M4) Additive inverse: For $x \in \mathbb{F} \setminus \{0\}$, $\exists x^{-1} \in \mathbb{F}$ s.t. $xx^{-1} = 1$

Theorem 1.4 (Distributive property)

- (D) $x(y + z) = xy + xz$ for all $x, y, z \in \mathbb{F}$.

Definition 1.5 (Field)

A set $\mathbb{F}$ which defines the following two operations:
- Addition: an operation that maps $x, y \in \mathbb{F} \to c \in \mathbb{F}$ and satisfies the properties of addition
- Multiplication: an operation that maps $x, y \in \mathbb{F} \to c \in \mathbb{F}$ and satisfies the properties of multiplication

for which the distributive property also holds.

Theorem 1.6

$\mathbb{R}$ and $\mathbb{C}$ are fields.

Definition 1.7 (Complex number)

A number which can be defined by a pair of real numbers a, b where the value of the number is equal to $a + bi$.

Theorem 1.8 (Useful things about complex numbers)

Let $z = a + bi$ and $w = c + di$ be complex numbers. Then:
- $z + w = (a + c) + (b + d)i$
- $zw = (ac - bd) + (bc + ad)i$

2. Vector spaces

2.1. Vector spaces

Definition 2.1.1 (Vector space)

A vector space over $\mathbb{F}$ is a set V with the following operations:
- **Vector addition**: $u \in V, v \in V \mapsto (u + v) \in V$, which satisfies the properties of addition
- **Scalar multiplication**: $\alpha \in \mathbb{F}, v \in V \mapsto \alpha v \in V$, which satisfies the properties of scalar multiplication

Definition 2.1.2 (Properties of addition)

For all $u, v, w \in V$:
- (A1) Commutativity: $u + v = v + u$
- (A2) Associativity: $(u + v) + w = u + (v + w)$
- (A3) Identity: $\exists 0 \in \mathbb{R}$ s.t. $0 + u = u$
- (A4) Additive inverse: For $u \in \mathbb{F}$, $\exists -u \in \mathbb{F}$ s.t. $u + (-u) = 0$

Definition 2.1.3 (Properties of scalar multiplication)

For all $\alpha, \beta \in \mathbb{F}$, $v, w \in V$:
- (S1) Associativity: $(\alpha\beta)v = \alpha(\beta v)$
- (S2) Distributivity over scalar addition: $(\alpha + \beta)v = \alpha v + \beta v$
- (S3) Distributivity over vector addition: $\alpha(v + w) = \alpha v + \alpha w$
- (S4) Multiplicative identity: $1v = v$

Theorem 2.1.4

$\mathbb{F}^n$ is a vector space.

Theorem 2.1.5

All inverses and identities are unique in a vector space.

2.2. Subspaces

Definition 2.2.1 (Subspace)

Let V be any vector space, and let W be a subset of V. Define vector addition and scalar multiplication on W by restricting the corresponding operations of V to W. If W is a vector space with respect to the restricted operations of V, then W is said to be a subspace of V.

Definition 2.2.2 (Closed)

An operation is closed under a set if applying the operation to elements of the set always results in an element of the set.

Definition 2.2.3 (Subspace)

A subspace of a vector space V is a subset W of V which contains the zero vector and is closed under addition and scalar multiplication.

Theorem 2.2.4

A subset W of a vector space V is a subspace iff
- (i) W is nonempty
- (ii) $\alpha \in \mathbb{F}$ and $w_1, w_2 \in W$ implies $\alpha w_1 + w_2 \in W$

Typically, we prove (i) by proving that $0 \in W$.

2.3. Subspaces of $\mathbb{F}^n$

Theorem 2.3.1 (Subspaces of $\mathbb{R}^n$)

$\mathbb{R}^n$ contains the following subspaces:
- $\{0\}$
- $\mathbb{R}^n$
- Any line through the origin
- Any plane through the origin
- etc

Definition 2.3.2 (Spanning)

For vectors to span a space is for it to be sufficient to be able to reach any point in the space using the vectors.

Definition 2.3.3 (Independence)

If a vector can be made out of other vectors, then the vector is independent

2.4. Intersections and unions of subspaces

Theorem 2.4.1

The intersection of any collection of subspaces of V is a subspace of V.

Theorem 2.4.2

The union of two subspaces of V is a subspace of V iff one of the subspaces is contained in the other.

Definition 2.4.3 (Sum of subspaces)

If U and W are subspaces of a vector space V, then

$$U + W := \{u + w : u \in U \text{ and } w \in W\}$$

Theorem 2.4.4

If U and W are subspaces of V, then $U + W$ is the smallest subspace of V containing both U and W.

Definition 2.4.5 (Direct sum)

If $V_1, ..., V_n$ are subspaces of V such that each element of $\sum_{k=1}^{n} V_k = V_1 + \cdots + V_n$ can be written uniquely as $\sum_{k=1}^{n} v_k = v_1 + \cdots + v_n$ where $v_k \in V_k$, then $\sum_{k=1}^{n} V_k$ is a **direct sum** and can be written as $\bigoplus_{k=1}^{n} V_k = V_1 \oplus \cdots \oplus V_n$.

Theorem 2.4.6

Let $V_1, ..., V_n$ be subspaces of V. Then they are direct sums iff the only way to write $0 = v_1 + \cdots + v_n$ is to take $v_1 = \cdots = v_n = 0$.

Theorem 2.4.7

If U and W are subspaces of V, then $U + W$ is a direct sum iff $U \cap W = \{0\}$. This does not generalize to higher numbers of subspaces.

2.5. Spanning

Definition 2.5.1 (Linear combination)

A linear combination of a collection $v_1, ..., v_n$ of vectors in vector space V is a vector of the form

$$\alpha_1 v_1 + \cdots + \alpha_n v_n$$

where each $\alpha_k \in \mathbb{F}$.

Definition 2.5.2 (Span)

Given $W \subseteq V$ where V is a vector field, the set of all linear combinations of vectors in W is called the span of W.

$$\text{span}(W) := \left\{ \sum_{i=1}^{n} \alpha_i w_i : \alpha_i \in \mathbb{F}, w_i \in W \right\}$$

Additionally, we define

$$\text{span}(\varnothing) = \{0\}$$

Definition 2.5.3 (Subspace generated by a set)

Given $W \subseteq V$ where V is a vector field, the subspace generated by W is the smallest subspace of V containing W, or equivalently the intersection of all subspaces of V containing W.

Theorem 2.5.4

The span of W is the subspace generated by W, i.e. W is the smallest subspace of V containing W.

Definition 2.5.5 (Spanning, spanning set)

If $\text{span}(W) = V$, then W spans V and W is a spanning set for V.

Definition 2.5.6 (Finite-dimensional vector space)

A vector space is finite-dimensional iff it has a finite spanning set.

Otherwise, it is infinite-dimensional.

2.6. Linear independence

Definition 2.6.1 (Linear independence)

Let V be a vector space. If $W \subseteq V$ is a finite set, it is ***linearly independent*** iff one of the following equivalent conditions:

- the only way to write 0 as a combination

$$\alpha_1 v_1 + \cdots + \alpha_m v_m = 0$$

  is by taking $\alpha_1 = \cdots = \alpha_m = 0$.
- there does not exist $w \in W$ (possibly $w = 0$) s.t. w is a linear combination of the other elements of W.

We also define $\varnothing$ to be linearly independent.

If $W \subseteq V$ is an infinite set, it is linearly independent if fevery finite subset of W is linearly independent.

Linear dependence is the opposite of linear independence.

Theorem 2.6.2

If $W \subseteq V$ is linearly independent, any subset $U \subseteq W$ is linearly independent.

Theorem 2.6.3

Let A be an $m \times n$ matrix. The following are equivalent:
- The columns of A are linearly independent.
- $Ax = 0$ has only the solution $x = 0$.
- A has n pivots.
- there are no free variables for $Ax = 0$.

2.7. Basis

Definition 2.7.1 (Basis)

A basis of V is a subset of V which is linearly independent and spans V.

2.8. Dimension

Theorem 2.8.1 (Plus/minus lemma)

Let $S \subseteq V$ (where V is a vector space).
- If S is linearly independent, and v is not in the span of S, then $S \cup \{v\}$ is linearly independent.
- If $v \in \text{span}(S \setminus \{v\})$, then $\text{span}(S) = \text{span}(S \setminus \{v\})$.

Theorem 2.8.2

Let V be a finite-dimensional vector space and $S \subseteq V$. Then,
- If $\text{span}(S) = V$, then S contains a subset B which is a basis of V
- If S is linearly independent, then S can be extended to a basis of V

Theorem 2.8.3

Let V be a finite-dimensional vector space spanned by a set of m vectors. Then any linearly independent set of vectors in V is finite and contains no more than m elements.

2.9. Example vector spaces

Definition 2.9.1 ($\mathbb{F}^n$)

For a field $\mathbb{F}$, $\mathbb{F}^n$ is the set of all ordered n-tuples of elements of $\mathbb{F}$:

$$\mathbb{F}^n := \{(x_1, ..., x_n) : x_1, ..., x_n \in \mathbb{F}\}$$

where addition and scalar multiplication are defined as expected:

$$a + b := (a_1 + b_1, ..., a_n + b_n)$$
$$\alpha x := (\alpha x_1, ..., \alpha x_n)$$

This is an n-dimensional vector space over $\mathbb{F}$.

Its standard ordered basis is

$$\left(\begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, ..., \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix} \right).$$

The standard basis vectors are called $\hat{\imath}$, $\hat{\jmath}$, $\hat{k}$ for $n \leq 3$, and called $\hat{e}_1, ..., \hat{e}_n$ for any n.

Definition 2.9.2 ($\mathbb{F}^\omega$)

The set of all sequences of elements of $\mathbb{F}$:

$$\mathbb{F}^\omega := \{(x_1, x_2, ...) : x_k \in \mathbb{F} \text{ for } k \in \mathbb{N}\}$$

where addition and scalar multiplication are defined similarly to $\mathbb{F}^n$:

$$a + b := (a_1 + b_1, ..., a_n + b_n)$$
$$\alpha x := (\alpha x_1, ..., \alpha x_n)$$

This is

Definition 2.9.3 ($\mathbb{F}^{m \times n}$)

The set of all $m \times n$ matrices with entries in $\mathbb{F}$, where addition and scalar multiplication are defined as:

$$(A + B)_{ij} := A_{ij} + B_{ij}$$
$$(\alpha A)_{ij} := \alpha A_{ij}$$

$\mathbb{F}^{m \times n}$ is a vector space of dimension mn.

$\mathbb{F}^{m \times n}$ is also denoted $M_{m \times n}(\mathbb{F})$, $\text{Mat}(m, n; F)$, or similar notations.

Definition 2.9.4 (Vector space of functions)

Let V be a vector space, S be a set, and

$$V^S = \{f : S \to V\}$$

(the set of all functions that map members of S to members of V). Then V^S is a vector space, if we define for all $p, q \in V^S$, $s \in \mathbb{F}$,

$$(f + g)(s) = f(s) + g(s) \quad (\alpha f)(s) = \alpha(f(s))$$

Definition 2.9.5 (Polynomial)

A function $p : \mathbb{F} \to \mathbb{F}$ is a polynomial of degree n iff there exist $c_0, ...c_n \in \mathbb{F}$ such that

$$p(x) = c_0 + c_1 x + c_2 x^2 + \cdots c_n x^n = \sum_{k=0}^{n} c_k x^k$$

$\mathcal{P}(\mathbb{F})$ is the set of all polynomials of any degree with coefficients in $\mathbb{F}$. $\mathcal{P}_n(\mathbb{F})$ is the set of all polynomials of degree n with coefficients in $\mathbb{F}$.

$\mathcal{P}(\mathbb{F})$ and $\mathcal{P}_n(\mathbb{F})$ are vector spaces if we define for all $p, q \in \mathcal{P}_n(\mathbb{F})$, $s \in \mathbb{F}$,

$$(p+q)(s) = p(s) + q(s) \quad (\alpha p)(s) = \alpha(p(s))$$

3. Linear maps

3.1. Linear map

Definition 3.1.1 (Linear map)

Let V and W be vector spaces over $\mathbb{F}$. A **linear map** (also called **linear function** or **linear transformation**) from V to W is a function $T : V \to W$ with the two properties

$$T(v_1 + v_2) = Tv_1 + Tv_2 \quad \text{(additivity)}$$
$$T(\alpha v) = \alpha T v \quad \text{(homogeneity)}$$

Equivalently, it is a function with the property that

$$T(\alpha v_1 + v_2) = \alpha T v_1 + T v_2$$

Equivalently, it is a function with the property that

$$T(\alpha_1 v_1 + \alpha_2 v_2) = \alpha_1 T v_1 + \alpha_2 T v_2$$

Note: Not all elementary linear functions $y = mx + b$ are linear maps! All linear maps in $\mathbb{F}^1$ are of the form $y = mx$.

Definition 3.1.2 (Linear operator)

A function $T : V \to V$ which is a linear map.

Definition 3.1.3 ($\mathcal{L}$)

For any vector spaces V and W, $\mathcal{L}(V, W)$ is the set of all linear maps from V to W.

$\mathcal{L}(V) := \mathcal{L}(V, V)$.

$\mathcal{L}(V, W)$ is also called $\mathrm{Hom}(V, W)$.

Lemma 3.1.4 (Linear Map Lemma)

Let V be a finite-dimensional vector space and W be a vector space. Suppose $\{v_1, ...v_n\}$ is a basis of V and $w_1 ... w_n \in W$. Then there exists a unique linear map $T : V \to W$ such that

$$Tv_k = w_k$$

for each $k \in \{1, ..., n\}$.

Note: This means that there exists a unique linear map that maps a basis to any vectors we wish, and that a linear map is uniquely determined by its output on a basis.

Lemma 3.1.5

If $T : V \to W$ is a linear map, then $T(0) = 0$.

Theorem 3.1.6 ($\mathcal{L}(V, W)$ is a vector space)

If we define

$$(S+T)(v) := S + T$$
$$(\alpha T)(v) := \alpha T(v)$$

for $S, T \in \mathcal{L}(V, W)$, $\alpha \in \mathbb{F}$, then $\mathcal{L}(V, W)$ is a vector space and is a subspace of V^W.

Definition 3.1.7 (Product of linear maps)

If $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$, then we define the product $ST \in \mathcal{L}(V, W)$ by $ST := S \circ T$.

Theorem 3.1.8 ($\mathcal{L}(V)$ is a unital associative F-algebra)

The product of linear maps on V has the following properties:

- **Bilinearity**: For all $S, T_1, T_2 \in \mathcal{L}(V)$, $\alpha \in \mathbb{F}$,
  - $S(T_1 + T_2) = ST_1 + ST_2$
  - $(S_1 + S_2)T = S_1 T + S_2 T$
  - $(\alpha S)T = \alpha(ST) = S(\alpha T)$
- **Associativity**: $(RS)T = R(ST)$ for all $R, S, T \in \mathcal{L}(V)$
- **Identity** (*unital*): $IT = TI = T$ for all $T \in \mathcal{L}(V)$, where I is the identity map $x \mapsto x$.

It is an *F-algebra* because it is a vector space over a field equipped with bilinear multiplication.

3.2. Kernel

See Trimm 4.2, Karthik 4.2, Axler 3B

Definition 3.2.1 (Kernel)

The **kernel** or **null space** of a linear map $T : V \to W$ is the set of vectors in V which T maps to the zero vector of W:

$$\operatorname{null} T = \ker T := \{v \in V : Tv = 0\} \subseteq V$$

Theorem 3.2.2 (Kernel is subspace of domain)

If $T : V \to W$, then $\ker T$ is a subspace of V.

Definition 3.2.3 (Injective)

A map of sets $f : A \to B$ is **injective** or **one-to-one** iff $a_1 \neq a_2$ implies $f(a_1) \neq f(a_2)$, or equivalently iff $f(a_1) = f(a_2)$ implies $a_1 = a_2$.

Theorem 3.2.4 (Injective linear map has trivial kernel)

A linear map $T : V \to W$ is injective iff $\ker T = \{0\}$.

3.3. Image

See Trimm 4.2, Karthik 4.2, Axler 3B

Definition 3.3.1 (Image)

Let $T : V \to W$ be a linear map. The **image** or **range** of T is the set of all outputs of T:

$$\operatorname{range} T = \operatorname{im} T := \{Tv : v \in V\} \subseteq W$$

Theorem 3.3.2 (Image is subspace of codomain)

If $T : V \to W$, then $\operatorname{im} T$ is a subspace of W.

Definition 3.3.3 (Surjective)

A map of sets $f : A \to B$ is surjective iff $\operatorname{im} f = B$, or equivalently iff for every $b \in B$ there exists $a \in A$ s.t. $f(a) = b$.

Iff f is surjective, f **maps A onto B**.

3.4. Fundamental Theorem

Theorem 3.4.1 (Fundamental Theorem of Linear Maps)

Let $T : V \to W$ be a linear map with V finite-dimensional. Then $\operatorname{im} T$ is finite-dimensional and

$$\dim V = \dim \ker T + \dim \operatorname{im} T$$

Theorem 3.4.2 (Corollary to Fundamental Thm)

Let V and W be finite-dimensional vector spaces, and let $T : V \to W$. Then

- If $\dim V > \dim W$, then T is not injective.
- If $\dim V < \dim W$, then T is not surjective.

3.5. Invertible linear maps

3.6. Isomorphisms

Definition 3.6.1 (Bijective)

Injective and surjective.

Definition 3.6.2 (Isomorphism)

Let V and W be vector spaces over $\mathbb{F}$. An **isomorphism from V to W** is a bijective linear map $T : V \to W$. Iff there exists an isomorphism from V to W, V and W are **isomorphic**, which is denoted by $V \cong W$.

Definition 3.6.3 (Identity map)

The identity map $\mathrm{id}_V : V \to V$ is defined such that

$$\mathrm{id}_V(v) = v \quad \text{for } v \in V$$

Theorem 3.6.4 (Isomorphism is an equivalence relation)

- **Reflexive**: For any vector space V, $\mathrm{id}_V : V \to V$ is an isomorphism, i.e. $V \cong V$ for every V.
- **Symmetric**: If $T : V \to W$ is an isomorphism, then $T^{-1} : W \to V$ is also an isomorphism. Thus, $V \cong W$ implies $W \cong V$.
- **Transitive**: If $T_1 : U \to V$ and $T_2 : V \to W$ are isomorphisms, then $T_2T_1 : U \to W$ is an isomorphism. Thus, $U \cong V$ and $V \cong W$ implies $U \cong W$

Therefore, isomorphism is an equivalence relation on the collection of all vector spaces over $\mathbb{F}$.

Definition 3.6.5 (Isomorphism class)

The isomorphism class $[V]$ of a vector space V is the set of all vector spaces isomorphic to V.

Lemma 3.6.6

Any two isomorphism classes are disjoint or equal.

Theorem 3.6.7

Two finite-dimensional vector spaces over $\mathbb{F}$ are isomorphic iff they have the same dimension.

Equivalently, any finite-dimensional vector space V over $\mathbb{F}$ is isomorphic to $\mathbb{F}^{\dim V}$.

Theorem 3.6.8

If V and W are finite-dimensional vector spaces of the same dimension and $T : V \to W$ is a linear map, then if T is injective or surjective, it is an isomorphism.

Lemma 3.6.9

$\mathcal{P}_n(\mathbb{F}) \cong \mathbb{F}^{n+1}$.

4. Linear systems and matrix arithmetic

Having developed linear maps abstractly, we now fix bases and see how they correspond exactly to matrices, then specialize to $\mathbb{F}^n$, where the central object of study is a system $AX = Y$.

4.1. Matrices

See Axler [3C]

Definition 4.1.1

An $m \times n$ matrix is a rectangular array of numbers with m rows and n columns.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

a_{ij} is in the ith row and jth column of A.

Definition 4.1.2 (Matrix sum and scalar product)

Given $m \times n$-matrices A and B, their sum $A + B$ is

$$A + B := \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \dots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \dots & a_{2n} + b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \dots & a_{mn} + b_{mn} \end{bmatrix}.$$

Addition is only defined if A and B have the same number of rows and columns.

If $c \in \mathbb{F}$, the scalar product cA is

$$cA := \begin{bmatrix} ca_{11} & ca_{12} & \dots & ca_{1n} \\ ca_{21} & ca_{22} & \dots & ca_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ ca_{m1} & ca_{m2} & \dots & ca_{mn} \end{bmatrix}.$$

Definition 4.1.3 (Column, row vector)

A column vector is an $m \times 1$-matrix. A row vector is a $1 \times n$-matrix.

Definition 4.1.4 (Transpose of a matrix)

If A is an $m \times n$ matrix, then its transpose A^{T} is the $n \times m$ matrix obtained by interchanging the rows and columns of A:

$$\left[A^{\mathrm{T}}_{ij}\right] = \left[A_{ji}\right]$$

Remark

The transpose of a column vector is a row vector and vice versa.

4.2. Coordinates

See Trimm [4.5], Karthik [4.5]

Definition 4.2.1 (Ordered basis)

An ordered basis of a n-dimensional vector space V is an n-tuple which is an ordering of a basis of V.

For the following definitions, let $B := (v_1, \dots, v_n)$ be an ordered basis of V, where V is an n-dimensional vector space over $\mathbb{F}$.

Definition 4.2.2 (Linear combination map (basis isomorphism))

The **linear combination map** or **basis isomorphism** is the isomorphism $L_B : \mathbb{F}^n \to V$ defined by

$$L_B(\vec{x}) = x_1 v_1 + \dots + x_n v_n$$

Definition 4.2.3 (Coordinate isomorphism)

The **coordinate isomorphism** is the isomorphism $L_B^{-1} : V \to \mathbb{F}^n$.

This means that for any vector space, its vectors can be expressed as vectors in $\mathbb{F}^n$.

Definition 4.2.4 (Coordinate vector)

Let v be a vector in V. Then the coordinate vector $[v]_B \in \mathbb{F}^n$ of v is defined as

$$[v]_B := L_B^{-1}(v)$$

Equivalently, it is the vector such that

$$L_B([v]_B) = v$$

4.3. Matrix of a linear map

See Axler [3C], Trimm [4.5], Karthik [4.5]

Definition 4.3.1 (Matrix of a linear map)

Let $B_V = (v_1, \dots, v_n)$ be a basis of V and $B_W = (w_1, \dots, w_m)$ be a basis of W. Let $T : V \to W$ be a linear map.

Then the matrix of T with respect to B_V and B_W is denoted as

$$[T]_{B_W B_V}.$$

It is the $m \times n$ matrix defined such that its jth column is the coordinate vector of Tv_j with respect to B_W:

$$\left([T]_{B_W B_V}\right)_{*j} = \left[Tv_j\right]_{B_W}.$$

$$[T]_{B_W B_V} = \begin{bmatrix} Tv_0 & \cdots & Tv_n \end{bmatrix}_{B_W}$$

Equivalently, it is the $m \times n$ matrix that satisfies

$$Tv_j = \sum_{i=1}^{m} \left([T]_{B_W B_V}\right)_{ij} w_i.$$

Lemma 4.3.2

Applying a linear map T to a vector v is equivalent to multiplying the coordinate vector of v by the matrix of T:

$$[Tv] = [T][v]$$

Definition 4.3.3 (Standard matrix)

The matrix of a linear map $T : \mathbb{F}^n \to \mathbb{F}^m$ with respect to the standard bases of $\mathbb{F}^n$ and $\mathbb{F}^m$.

Theorem 4.3.4

Suppose $S, T \in \mathcal{L}(V, W)$. Then $[S + T] = [S] + [T]$.

Suppose $\lambda \in \mathbb{F}$ and $T \in \mathcal{L}(V, W)$. Then, $[\lambda T] = \lambda[T]$.

Remark

Every $m \times n$ matrix A is the standard matrix of the linear map $T_A : \mathbb{F}^n \to \mathbb{F}^m$, $T_A(X) := AX$ (matrix-vector multiplication is defined precisely in Definition 4.5.1 below). This is the correspondence that lets us study the abstract map T_A purely as concrete array arithmetic on A, and it's the reason $\ker T_A$ (equivalently, the null space $\mathrm{Nul}\, A$) is

exactly the solution set of the homogeneous system $AX = 0$.

4.4. Linear systems

See MATH 257 [Module 1], Trimm [1.1, 1.2, 1.3]

Definition 4.4.1 (Linear equation)

Let V and W be vector spaces over $\mathbb{F}$, let $T : V \to W$ be a linear map, and let $y \in W$. A ***linear equation*** in the unknown $v \in V$ is an equation of the form

$$Tv = y.$$

When $V = \mathbb{F}^n$ and $W = \mathbb{F}$ (so T is a linear functional on $\mathbb{F}^n$), writing the standard matrix of T (Definition 4.3.3) as the row $[a_1 \quad \cdots \quad a_n]$ and $v = (x_1, \ldots, x_n)$, this recovers the familiar scalar form

$$\sum_k a_k x_k = y.$$

A ***solution*** to $Tv = y$ is an element $s \in V$ such that $Ts = y$.

Definition 4.4.2 (Linear system)

A ***linear system*** is a finite collection of linear equations

$$T_1 v = y_1, \quad \ldots, \quad T_m v = y_m$$

sharing the same unknown $v \in V$, where each $T_j : V \to W_j$ is a linear map and each $y_j \in W_j$.

A ***solution*** to the system is an $s \in V$ that solves every equation simultaneously, i.e. $T_j s = y_j$ for all j.

Theorem 4.4.3 (Combining linear maps into one)

Let $T_k : V \to W_k$ be linear maps for $k = 1, \ldots, m$. Then

$$T : V \to W_1 \times \cdots \times W_m, \quad Tv := (T_1 v, \ldots, T_m v)$$

is linear, and for any $Y = (y_1, \ldots, y_m) \in W_1 \times \cdots \times W_m$,

$$Tv = Y \quad \text{iff} \quad T_k v = y_k \text{ for every } k.$$

Corollary 4.4.3.1 (A linear system is a single linear equation)

A linear system $T_1 v = y_1, \ldots, T_m v = y_m$ has the same solution set as the single linear equation

$$Tv = Y$$

where $T : V \to W_1 \times \cdots \times W_m$ is defined by $Tv := (T_1 v, \ldots, T_m v)$ and $Y := (y_1, \ldots, y_m) \in W_1 \times \cdots \times W_m$.

Remark

When $V = \mathbb{F}^n$ and all $W_k = \mathbb{F}$, this is exactly how a linear system becomes the matrix equation $AX = Y$ — the standard matrix of T is the $m \times n$ matrix A whose jth row is the standard matrix of T_j — i.e. gluing the equations' coefficient rows together. Concretely, if m is the number of equations and n the number of variables, the jth equation

$$\sum_{k=1}^{n} A_{jk} X_k = Y_j$$

glues into

$$A = \begin{bmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & \ddots & \vdots \\ A_{m1} & \cdots & A_{mn} \end{bmatrix} \quad X = \begin{bmatrix} X_1 \\ \vdots \\ X_n \end{bmatrix} \quad Y = \begin{bmatrix} Y_1 \\ \vdots \\ Y_m \end{bmatrix},$$

giving the compact form $AX = Y$. (Matrix-vector multiplication, which makes precise sense of AX, is defined in Definition 4.5.1 below.)

Remark

This is the "row" side of the same picture whose "column" side is Definition 4.5.1: gluing m scalar equations $T_j v = y_j$ **row-by-row** into A builds the map $T = (T_1, \ldots, T_m) : V \to \mathbb{F}^m$ above, while — as the next section shows — reading A's **columns** as vectors and taking AX to mean the linear combination $X_1 a_1 + \cdots + X_n a_n$ recovers the same equation $AX = Y$ from the span perspective. One matrix, two equivalent readings.

Definition 4.4.4 (Coefficient and augmented matrix)

Given a linear system with combined equation $TX = Y$ (equivalently $AX = Y$) and $V = \mathbb{F}^n$, its ***coefficient*** matrix is the standard matrix A of T, and its ***augmented*** matrix is $[A \mid Y]$.

Definition 4.4.5 (Consistent linear system)

A system that has at least one solution.

Definition 4.4.6 (Equivalent linear systems)

Two systems are equivalent iff they have the same set of solutions.

4.5. Matrix-vector multiplication

See MATH 257 [Module 5, Module 6], Chuang [Lecture 3]

See 3Blue1Brown [3]

Remark

Consider $T : \mathbb{F}^2 \to \mathbb{F}^2$. Since $[T] = [T\hat{\imath} \;\; T\hat{\jmath}]$, and

$$\begin{aligned} T(x,y) &= T((x,0)+(0,y)) \\ &= T(x,0)+T(0,y) \\ &= T(x\hat{\imath})+T(y\hat{\jmath}) \\ &= xT(\hat{\imath})+yT(\hat{\jmath}), \end{aligned}$$

we define $A\vec{x} \coloneqq x_1 A_{*1} + x_2 A_{*2}$. The following definition generalizes this.

Definition 4.5.1 (Matrix-vector multiplication)

Let $\vec{x} \in \mathbb{F}^n$ and $A = [A_{*1} \;\; \cdots \;\; A_{*n}]$ be an $m \times n$-matrix. We define the product AX by

$$A\vec{x} \coloneqq x_1 A_{*1} + x_2 A_{*2} + \cdots + x_n A_{*n}.$$

- $A\vec{x}$ is a linear combination of the columns of A using the entries in $\vec{x}$ as coefficients.
- $A\vec{x}$ is only defined if the number of entries of $\vec{x}$ is equal to the number of columns of A.

Remark

Solving linear systems is the same as finding linear combinations: solving $A\vec{x} = \vec{y}$ means finding coefficients expressing Y as a linear combination of A's columns, i.e. finding $Y \in \text{span}\{a_1, ..., a_n\}$ (span: see [Vector spaces]).

Theorem 4.5.2

Let $A = [\vec{a}_1 \;\; \cdots \;\; \vec{a}_n]$ be an $m \times n$ matrix and $\vec{y} \in \mathbb{F}^m$. For any $(x_1, ..., x_n) \in \mathbb{F}^n$, the following are equivalent:

- $(x_1, ..., x_n)$ is a solution of the vector equation $x_1\vec{a}_1 + \cdots + x_n\vec{a}_n = \vec{y}$
- $\vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ is a solution of the matrix equation $A\vec{x} = \vec{y}$
- $(x_1, ..., x_n)$ is a solution of the linear system with augmented matrix $[A \mid \vec{y}]$

Corollary 4.5.2.1

The equation $A\vec{x} = \vec{y}$ has a solution iff $\vec{y} \in \text{span}\{\vec{a}_1, ..., \vec{a}_n\}$.

4.6. Row operations

See MATH 257 [Module 2], Trimm [1.3]

Definition 4.6.1 (Elementary row operations)

The elementary row operations are:

- **Scaling:** $R_i \mapsto cR_i$ where c is a nonzero scalar.
- **Replacement:** $R_i \mapsto R_i + cR_j$ where c is a scalar.
- **Interchange:** Swap R_i and R_j.

Theorem 4.6.2 (Elementary row operations are invertible)

For any elementary row operation e, there exists an elementary row operation e^{-1} such that $e^{-1}(e(A)) = A$ for any matrix A.

Definition 4.6.3 (Row equivalence)

Two matrices are row-equivalent iff each can be derived from the other using a finite number of elementary row operations.

Remark

Each elementary row operation replaces an equation of the system by a linear combination of itself with (or a scalar multiple of, or a swap with) another equation — this is why row-equivalent augmented matrices give equivalent systems.

Theorem 4.6.4

If the augmented matrices of two linear systems are row equivalent, then the two systems are equivalent.

4.7. Echelon forms

See MATH 257 [Lecture 3, 4], Trimm [1.4]

Definition 4.7.1 (Row echelon form (REF))

A matrix is in REF iff it satisfies:

1. All nonzero rows are above all rows of all zeros.
2. Each leading entry of a row is in a column to the right of the leading entry of the row above it.
3. All entries in a column below a leading entry are zeros.

Definition 4.7.2 (Reduced row echelon form (RREF))

A matrix is in RREF iff it is in REF and additionally satisfies:

4. The leading entry in each nonzero row is 1.
5. Each leading 1 is the only nonzero entry in its column

Definition 4.7.3

$\text{RREF}(A)$ is the unique matrix which is in RREF and is row-equivalent to A.

Definition 4.7.4 (Pivot position)

A location A_{ij} where $\text{RREF}(A)_{ij}$ is a leading 1.

Definition 4.7.5 (Pivot column)

A column which contains a pivot position.

Definition 4.7.6 (Pivot)

A nonzero number at a pivot position.

Definition 4.7.7 (Leading variable, determined variable, basic variable, pivot variable)

A variable in a pivot column.

Definition 4.7.8 (Free variable)

A variable not in a pivot column.

Procedure 4.7.9 (Gauss-Jordan elimination)

Procedure 4.7.10 (Gaussian elimination)

Iterate through the pivot columns of A from left to right.

For each pivot column, use elementary row operations to ensure that:

- the pivot position is nonzero
- all entries in the column below the pivot position are zero

This produces $\text{REF}(A)$.

Procedure 4.7.11 (Jordan elimination)

Iterate through the pivot columns of $\text{REF}(A)$ from right to left. For each pivot column, use elementary row operations to ensure that:

- all other entries in the column other than the pivot are zero
- the pivot is equal to 1.

This produces $\text{RREF}(A)$.

4.8. Homogeneous linear systems

Definition 4.8.1 (Homogeneous linear system)

A system where $y_0 = y_1 = \cdots = y_m = 0$. It can be written as $AX = 0$.

Theorem 4.8.2 (Trivial solution)

For any homogeneous system, $x_0 = x_1 = \cdots = x_n = 0$ is a solution to the system. Therefore, all homogeneous systems are consistent.

Theorem 4.8.3

- If there are less equations than there are variables ($m < n$), then $AX = 0$ has an infinite number of solutions.
- If there are an equal number of equations and variables, then A is row-equivalent to the $n \times n$ identity iff $AX = 0$ has only the trivial solution.
- If there are more equations than there are variables ($m > n$), then

$$\text{RREF}(A) = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \\ 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

iff $AX = 0$ has only the trivial solution.

Procedure 4.8.4 (Solution)

To solve a homogeneous system, perform Gauss-Jordan elimination on A so that $R = \text{RREF}(A)$. Then solve $RX = 0$. The variables which are not in pivot columns are free variables and may be set to any value, typically denoted $u_1, u_2, \ldots$.

4.9. Inhomogeneous linear systems

Procedure 4.9.1 (Solution)

To solve an inhomogeneous system, perform Gauss-Jordan elimination on $A' = [A \mid Y]$ so that $R' = [R \mid Z] = \text{RREF}(A')$. Then solve $RX = Z$. Note that not all inhomogeneous systems are solvable (consistent).

Theorem 4.9.2

If A is $m \times n$ with more equations than variables ($m > n$), then there exists $Y \in \mathbb{F}^m$ such that $AX = Y$ is inconsistent.

4.10. Matrix multiplication

See Axler [3C], MATH 257 [Module 7, 8]

Remark

Let $T_1 \in \mathcal{L}(\mathbb{F}^n, \mathbb{F}^m)$, $T_2 \in \mathcal{L}(\mathbb{F}^p, \mathbb{F}^n)$. We want to determine $[T_1 \circ T_2]$ in terms of $[T_1]$ and $[T_2]$.

Recall that the columns of the matrix of a linear map are the coordinate vectors of the map applied to each basis vector:

$$A := [T_1] = \begin{bmatrix} [T_1 e_1] & \cdots & [T_1 e_n] \end{bmatrix}$$
$$B := [T_2] = \begin{bmatrix} [T_2 e_1] & \cdots & [T_2 e_p] \end{bmatrix}$$

(**Note:** The number of columns is the input dimension, and the number of rows is the output dimension.)

Thus, we need to find $\left[(T_1 \circ T_2)(e_j)\right]$ for each $j \in \{1, \ldots, p\}$.

$$\begin{aligned} \left[(T_1 \circ T_2)(e_j)\right] &= \left[T_1(T_2 e_j)\right] \\ &= [T_1][T_2 e_j] \\ &= AB_{*j}. \end{aligned}$$

Since this holds for every column j, $[T_1 \circ T_2] = \begin{bmatrix} AB_{*1} & \cdots & AB_{*p} \end{bmatrix}$, so we define AB this way.

Definition 4.10.1 (Matrix multiplication)

Let A be an $m \times n$-matrix and let $B = \begin{bmatrix} B_{*1} & \cdots & B_{*p} \end{bmatrix}$ be an $n \times p$-matrix. We define

$$AB := \begin{bmatrix} AB_{*1} & AB_{*2} & \dots & AB_{*p} \end{bmatrix}$$

Theorem 4.10.2

Let A be an $m \times n$ matrix and B be an $n \times p$ matrix. Then for every $x \in \mathbb{F}^p$

$$A(Bx) = (AB)x.$$

Lemma 4.10.3

Composing two linear maps T_1 and T_2 is equivalent to multiplying the matrices of the two linear maps:

$$[T_1 \circ T_2] = [T_1][T_2]$$

Note. Other ways to compute AB.

Procedure 4.10.4 (Row-Column Rule)

Let A be $m \times n$ and B be $n \times p$. Then, AB is formed by dot-producting the rows of the first (A) with the columns of the second (B):

$$AB = \begin{bmatrix} A_{1*}B_{*1} & \dots & A_{1*}B_{*p} \\ A_{2*}B_{*p} & \dots & A_{2*}B_{*p} \\ \vdots & \ddots & \vdots \\ A_{m*}B_{*p} & \dots & A_{m*}B_{*p} \end{bmatrix}$$

$$(AB)_{ij} = A_{i*}B_{*j}.$$

Procedure 4.10.5 (Outer Product Rule)

Let A be $m \times n$ and B be $n \times p$ such that

$$A = \begin{bmatrix} C_1 & \cdots & C_n \end{bmatrix}, \quad \text{and} \quad B = \begin{bmatrix} R_1 \\ \vdots \\ R_n \end{bmatrix}.$$

Then

$$AB = C_1R_1 + \cdots + C_nR_n$$

Definition 4.10.6

The identity matrix I_n of size n is defined as

$$I_n = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}.$$

Theorem 4.10.7

Let A be an $m \times n$ matrix and let B and C be matrices for which the indicated sums and products are defined.

- **Associative:** $A(BC) = (AB)C$
- **Distributive over matrices:** $A(B + C) = AB + AC$, $(B + C)A = BA + CA$
- **Distributive over scalars:** $r(AB) = (rA)B = A(rB)$ for every scalar r,
- **Linear:** $A(rB + sC) = rAB + sAC$ for every scalars r, s
- **Identity:** $I_mA = A = AI_n$

Theorem 4.10.8

The transpose of a product is the product of transposes in opposite order:

$$(AB)^{\mathrm{T}} = B^{\mathrm{T}}A^{\mathrm{T}}$$

Definition 4.10.9

Let A be a square matrix. We write A^k for $A \cdots A$, k-times; that is A^k is obtained by multiplying A k-times with itself.

5. Matrices as linear maps

5.1. Elementary matrices

See Trimm [4.7], Karthik [4.7], MATH 257 [Module 9]

Definition 5.1.1

An elementary matrix is one that is obtained by performing a single elementary row operation on an identity matrix.

A permutation matrix is one that is obtained by performing row exchanges on an identity matrix.

Theorem 5.1.2

If $e : \mathbb{F}^{n \times n} \to \mathbb{F}^{n \times n}$ is an elementary row operation and $A \in \mathbb{F}^{n \times n}$, then

$$e(A) = e(I)A.$$

Corollary 5.1.2.1

Let A and B be $m \times n$ matrices. Then, B is row-equivalent to A iff $B = e_k(I) \cdots e_1(I)A$, where $e_1(I), \dots, e_k(I)$ are elementary matrices of size $m \times m$,

5.2. Invertible matrices

See Trimm [4.7], Karthik [4.7], MATH 257 [Module 10, 11]

Remark

The inverse of a real number a is denoted by a^{-1}. For example, $7^{-1} = 1/7$ and $7 \cdot 7^{-1} = 7^{-1} \cdot 7 = 1$. Note that not all real numbers have an inverse. Namely, 0^{-1} is not defined as there is no real number b such that $0 \cdot b = 1$.

Definition 5.2.1 (Invertible matrix)

An $n \times n$ matrix A is ***invertible*** iff there is an $n \times n$ matrix A^{-1} satisfying

$$A^{-1}A = AA^{-1} = I_n$$

where I_n is the $n \times n$ identity matrix. There is a unique such B, called the inverse of A.

A matrix that is not invertible is ***singular***.

Lemma 5.2.2

The matrix of an isomorphism of finite-dimensional vector spaces is invertible.

Theorem 5.2.3

Suppose A and B are invertible. Then:
- A^{-1} is invertible and $(A^{-1})^{-1} = A$.
- AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$. More generally, any finite product $A_1 \cdots A_k$ of invertible $n \times n$ matrices is invertible, with $(A_1 \cdots A_k)^{-1} = A_k^{-1} \cdots A_1^{-1}$.
- A^{T} is invertible and $(A^{\mathrm{T}})^{-1} = (A^{-1})^{\mathrm{T}}$.

Theorem 5.2.4

Let A be an $n \times n$ matrix. Then $AX = Y$ has a unique solution for every $Y \in \mathbb{F}^n$ iff A is invertible, in which case the unique solution is $X = A^{-1}Y$.

Theorem 5.2.5

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $ad - bc \neq 0$, then A is invertible and

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

If $ad - bc = 0$, then A is not invertible.

Theorem 5.2.6 (Invertible Matrix Theorem)

If A is an $n \times n$ matrix, and is the matrix of the linear map T_A, the following conditions are equivalent:
- A is invertible.
- T_A is injective.
  - $A\vec{x} = \vec{y}$ has a unique solution for every $\vec{y} \in \mathbb{F}^n$.
  - $A\vec{x} = 0$ has only the trivial solution ($\vec{x} = \vec{0}$).
  - $\operatorname{Nul} A = \{0\}$
  - nullity $A = 0$
  - The columns of A are linearly independent.
- T_A is surjective.
  - $\operatorname{Col} A = \mathbb{F}^n$.
  - The columns of A span $\mathbb{F}^n$.
  - $\operatorname{rank} A = n$.
- T_A is bijective.
  - T_A is an isomorphism.
- A is row-equivalent to the $n \times n$ identity matrix.
  - A has pivots in every row and column.
  - A is a product of elementary matrices.
- $\det A \neq 0$.
- 0 is not an eigenvalue of A.

Theorem 5.2.7

Suppose A is invertible. Then every sequence of elementary row operations that reduces A to I_n also transforms I_n to A^{-1}. Equivalently, $[A \mid I]$ is row-equivalent to $[I \mid A^{-1}]$.

Procedure 5.2.8 (Computation of A^{-1})

Row-reduce the augmented matrix $[A \mid I]$. If A is row-equivalent to I, then $\operatorname{RREF}([A \mid I]) = [I \mid A^{-1}]$. Otherwise, A doesn't have an inverse.

Note – The following is not in any of the sources; it comes from the solutions to MATH 257 HW3.

Lemma 5.2.9 (Inverse elementary row operations)

The inverses of the elementary row operations are:

Forward	Inverse
$R_i \mapsto cR_i$	$R_i \mapsto \frac{1}{c}R_i$
$R_i \mapsto R_i + cR_j$	$R_i \mapsto R_i - cR_j$
$R_i \leftrightarrow R_j$	$R_i \leftrightarrow R_j$

Assume $c \neq 0$ and $i \neq j$.

Procedure 5.2.10 (Computation of inverse of matrix under elementary ops)

Let A be the result of applying $e_1, \ldots, e_n$ in order to I. Then, A^{-1} is the result of applying $(e_n)^{-1}, \ldots, (e_1)^{-1}$ in order to I.

Warning: The order must be reversed!

5.3. Left and right inverse

See MIT 18.06SC [Lecture 33], Strang-ILA-6e [4.5], MATH 257 [HW3.4], https://math.stackexchange.com/questions/1694351/finding-all-left-inverses-of-a-matrix

Let A be an $m \times n$ matrix with $n > m$. We wish to find B s.t. $AB = I$, i.e. B is the ***right-inverse*** of A. The right-inverse is not unique.

Theorem 5.3.1 (Finding left/right-inverse)

Notice that AA^{T} is square, and that

$$A\left(A^{\mathrm{T}}(AA^{\mathrm{T}})^{-1}\right) = (AA^{\mathrm{T}})(AA^{\mathrm{T}})^{-1} = -1.$$

Thus, a right-inverse of A is

$$A_{\text{right}}^{-1} := A^{\mathrm{T}}(AA^{\mathrm{T}})^{-1}.$$

It only exists when A has independent rows.

Similarly, we can find that a left-inverse of A is

$$A_{\text{left}}^{-1} := (A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}.$$

It only exists when A has independent columns.

Here are some alternative ways to do it. They aren't very useful.

Procedure 5.3.2 (Computation of right-inverse by zeroing out rows)

Choose $n - m$ rows of B to zero out. By the outer product rule, zeroing the jth row of B causes the jth column of A to contribute nothing to AB, so that

column can be ignored. After zeroing those rows, the remaining rows of B form a square matrix B', and the corresponding columns of A form a square matrix A'. For $AB = I$ to hold, we need $A'B' = I$, so $B' = (A')^{-1}$.

If A' is singular, choose a different set of rows to zero out. If no choice yields an invertible A', then no right-inverse exists.

Procedure 5.3.3 (Column-by-column computation of right-inverse)

Apply A to each column of B separately and set the result equal to the corresponding column of I, giving one linear system per column of B to solve.

5.4. LU decomposition

See MATH 257 [Module 12, 13], Strang-ILA-6e [2.3]

See (extras) Strang-LALD [I.4], Strang-ILA-5e [2.6], Strang-LAFE [2.3], Strang-ZoomNotes [2.3, 2.5], MIT 18.06SC [Lecture 4]

Definition 5.4.1

An $n \times n$ matrix A is called
- upper triangular iff it is of the form

$$\begin{bmatrix} \star & \star & \star & \star & \star \\ 0 & \star & \star & \star & \star \\ 0 & 0 & \star & \star & \star \\ 0 & 0 & 0 & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \star \end{bmatrix},$$

  i.e. all the entries below the main diagonal are zero.
- lower triangular iff it is of the form

$$\begin{bmatrix} \star & 0 & 0 & 0 & 0 \\ \star & \star & 0 & 0 & 0 \\ \star & \star & \star & 0 & 0 \\ \star & \star & \star & \ddots & \vdots \\ \star & \star & \star & \star & \star \end{bmatrix},$$

  i.e. all the entries above the main diagonal are zero.
- diagonal iff it only has entries on the main diagonal.

Theorem 5.4.2

The product of two lower (upper) triangular matrices is lower (upper) triangular.

The inverse of a lower (upper) triangular matrix (if it exists) is again lower (upper) triangular.

Definition 5.4.3 (LU decomposition)

A matrix A has an ***LU decomposition*** iff there is a lower triangular matrix L and an upper triangular matrix U such that $A = LU$.

Lemma 5.4.4

Not every matrix has an LU decomposition. The LU decomposition of a matrix is not unique.

Procedure 5.4.5 (Finding LU decomposition ($O(n^3)$))

Suppose that $\mathrm{REF}(A)$ can be found by operations $e_1, \ldots, e_n$ of the form $R_i \mapsto R_i + cR_j$ where $j < i$:

$$U := \mathrm{REF}(A) = e_n(I) \cdots e_1(I) A.$$

We define $U := \mathrm{REF}(A)$ since $\mathrm{REF}(A)$ is always upper-triangular. Finding U takes approximately $\frac{1}{3}n^3$ multiplications and $\frac{1}{3}n^3$ subtractions.

Then,

$$L := e_1^{-1}(I) \cdots e_n^{-1}(I)$$

is lower-triangular.

Lemma 5.4.6

The elementary matrix for a replacement ($R_i \mapsto R_i + cR_j$) with $j < i$ or scaling ($R_i \mapsto cR_i$) operation is lower triangular.

Theorem 5.4.7

Let A be an $n \times n$-matrix. If A can be brought to echelon form just using row operations of the form $R_i \mapsto R_i + cR_j$ where $j < i$, then A has an LU decomposition.

Lemma 5.4.8

Every matrix A has some permutation matrix (possibly the identity) s.t. $PA = LU$, where LU is an LU decomposition.

Equivalently, every matrix has some permutation of its rows which has an LU decomposition.

Procedure 5.4.9 (Using LU decomposition ($O(n^2)$))

Suppose $A = LU$ where L is lower triangular and U is upper triangular. Then, we can solve the problem $A\vec{x} = \vec{b}$ by:
1. Solving $L\vec{y} = \vec{b}$ for $\vec{y}$ using forward-substitution (going from top down).
2. Solving $U\vec{x} = \vec{y}$ for $\vec{x}$ using back-substitution (going from bottom up).

Remark

LU decomposition is most useful when one has to solve many linear systems $A\vec{x} = \vec{y}$ with the same coefficients but different right-hand side $\vec{y}$.

5.5. Fundamental matrix spaces

See MATH 257 [Module 17, Module 21], Trimm 4.6, Karthik 4.2 Axler 3C.31

Given a matrix $A \in \mathbb{F}^{m,n}$, define $T_A := x \mapsto Ax$. Then $T_A : \mathbb{F}^n \to \mathbb{F}^m$.

Definition 5.5.1 (Null space, nullity)

The **null space** of A is the kernel of T_A:

$$\operatorname{Nul} A := \ker T_A = \{x \in \mathbb{F}^n : Ax = 0\}$$

which is the solution set of the homogeneous linear system $Ax = 0$.

$$\operatorname{nullity} A := \dim \operatorname{Nul} A$$

Definition 5.5.2 (Column space, rank)

The **column space** of A is the image of T_A:

$$\begin{aligned}\operatorname{Col} A := \operatorname{im} T_A &= \{T_A x : x \in \mathbb{F}^n\} \\ &= \{Ax : x \in \mathbb{F}^n\} \\ &= \left\{\sum_{i=1}^{n} x_i A_{*i} : x_i \in \mathbb{F}\right\} \\ &= \operatorname{span}\{A_{*1}, \ldots, A_{*n}\}\end{aligned}$$

which is the span of the column vectors of the matrix A.

$$\operatorname{rank} A := \dim \operatorname{Col} A$$

Theorem 5.5.3 (Rank-nullity theorem)

$$\operatorname{rank} A = \text{the number of pivot columns}$$
$$\operatorname{nullity} A = \text{the number of non-pivot columns}$$

Therefore,

$$n = \operatorname{nullity} A + \operatorname{rank} A,$$

where n is the number of columns of A, or equivalently the dimension of the domain of T_A.

Procedure 5.5.4 (Finding bases of Nul A, Col A, Row A)

1. The pivot columns of A are a basis of Col A.
2. The pivot rows of A or of $\operatorname{RREF}(A)$ are a basis of Row A. (row space is preserved under row operations)
3. The vectors spanning $Ax = 0$ are a basis of Nul A, so
   1. Find the parametric form of the solutions to $Ax = 0$.
   2. Express solutions for x as a linear combination of vectors with the free variables as coefficients.
   3. These vectors form a basis of Nul A.

Definition 4.1.4 (Transpose of a matrix)

If A is an $m \times n$ matrix, then its transpose A^{T} is the $n \times m$ matrix obtained by interchanging the rows and columns of A:

$$\left[A^{\mathrm{T}}_{ij}\right] = \left[A_{ji}\right]$$

Definition 5.5.5 (Row space)

The row space of A is the span of the row vectors of A:

$$\operatorname{Row} A := \operatorname{span}\{A_{1*}, \ldots, A_{m*}\} = \operatorname{Col} A^T$$

Lemma 5.5.6

$$\dim \operatorname{Row} A = \dim \operatorname{Col} A = \operatorname{rank} A$$

Definition 5.5.7 (Left null space)

The left null space of A is the null space of A^T. It is the solution set of the homogeneous linear system $A^T y = 0$ or equivalently of $y^T A = 0^T$.

Procedure 5.5.8 (Finding the left null space without column operations)

To find the left null space of the $n \times m$ matrix A, make a column vector $b = (b_1, \ldots, b_m)^T$ and calculate $\operatorname{RREF}([A \mid b])$. Then the basis of the left null space is given by the rows of $\operatorname{RREF}([A \mid b])$ for which the left side is all zero (where $b_1, \ldots, b_m$ are the basis in which A is expressed in (the basis B for which $A = [T]_B$, which is usually the standard basis)).

Alternately, calculate $\operatorname{RREF}([A \mid I])$, then the basis of the left null space is given by the rows of $\operatorname{RREF}([A \mid I])$ for which the left side is all zero.

Lemma 5.5.9

$$\operatorname{Nul}(A^{\mathrm{T}}) = \operatorname{Col}(A)^{\perp}$$
$$\operatorname{Row}(A) = \operatorname{Col}(A^{\mathrm{T}}) = \operatorname{Nul}(A)^{\perp}$$

5.6. Change of basis

Definition 5.6.1 (Change of basis matrix)

The **change of basis matrix** or **transition matrix** of an n-dimensional vector space V with relation to the ordered bases B and B' is the $n \times n$ matrix whose jth column is the coordinate vector of the jth vector in B with respect to B':

$$C_{*j} := \left[v_j\right]_{B'}$$

Procedure 5.6.2 (Finding the transition matrix)

Given bases B and B', row-reduce the augmented matrix $[[w_i] \mid [v_i]]$, whose first n columns are the coordinate vectors $[w_i]$ of the vectors of B' and whose last n columns are the coordinate vectors $[v_i]$ of the vectors of B. Then the transition matrix from B to B' is the right-hand side of $\operatorname{RREF}([[w_i] \mid [v_i]])$.

Lemma 5.6.3 (Inverting a transition matrix)

$$C_{B' \to B} = C_{B \to B'}^{-1}$$

Note that $C_{B' \to B}$ is the matrix formed by the coordinate vectors of the basis vectors of B' with respect to B.

Lemma 5.6.4 (Changing the basis of a vector)

If B and B' are bases of V, $v \in V$, and $C_{B \to B'}$ is the transition matrix from B to B', then

$$[v]_{B'} = C_{B \to B'}[v]_B$$

Lemma 5.6.5 (Changing the basis of a linear map by changing its matrix)

If B and B' are bases of V, $T \in \mathcal{L}(V)$, and $C_{B \to B'}$ is the transition matrix from B to B', then

$$[T]_{B'} = C_{B \to B'}[T]_B C_{B' \to B}$$

Procedure 5.6.6 (Changing the basis of a linear map by applying the linear map to basis vectors)

If B and $B' = (w_1, ..., w_n)$ are bases of V, $T \in \mathcal{L}(V)$, and $C_{B \to B'}$ is the transition matrix from B to B', then

$$T_{B'} = \begin{bmatrix} C_{B \to B'}[T]_B[w_1]_B & \cdots & C_{B \to B'}[T]_B[w_n]_B \end{bmatrix}$$

(Apply T to each of the basis vectors of B' and then convert Tw_i to the basis B')

5.7. Similarity

Definition 5.7.1 (Similar matrices)

Two $n \times n$ matrices A and B are similar if there exists an invertible matrix C s.t. $B = C^{-1}AC$.

Similarity is an equivalence relation on $\mathbb{F}^{n,n}$.

Definition 5.7.2 (Similarity invariant)

A property of a $n \times n$ matrix A which holds for all matrices similar to A.

Theorem 5.7.3 (Useful similarity invariants)

The following properties are similarity invariants:
1. invertibility
2. nullity
3. rank

5.8. Linear functionals

Definition 5.8.1 (Linear functional)

Let V be a vector space over $\mathbb{F}$. Then a linear map $T : V \to \mathbb{F}$ is a **linear functional** on V.

Definition 5.8.2 (Dual space)

Let V be a vector space over $\mathbb{F}$. Then the dual space of V is the vector space of linear functionals on V, $V^* := \mathcal{L}(V, \mathbb{F})$.

Definition 5.8.3 (Kronecker delta)

$$\delta_{ij} := \begin{cases} 1 \text{ if } i = j \\ 0 \text{ if } i \neq j \end{cases}$$

Definition 5.8.4 (*i*th coordinate function)

Definition 5.8.5 (Dual basis)

Lemma 5.8.6

Let V be an n-dimensional vector space and $B = \{v_1, ...v_n\}$ be a basis of V. Then for each $i \in \{1, ..., n\}$ there exists a unique linear functional f_i (called the ***i*th coordinate function for B**) such that

$$f_i(v_j) = \delta_{ij} \quad \text{for all } j \in \{1, ..., n\}$$

For any $v \in V$, we can express v as a linear combination of elements of B, where the coefficients are given by $f_i(v)$:

$$v = \sum_{j=1}^{n} f_j(v) v_j$$

Additionally,

$$B^* := \{f_1, ..., f_n\}$$

is a basis of V^*, and is called the **dual basis** of the basis B of V.

Any linear functional $f : V \to \mathbb{F}$ can be written uniquely as

$$f = \sum_{i=1}^{n} f(v_i) f_i$$

and $f(v)$ can be written as

$$f(v) = \sum_{i=1}^{n} f(v_i)([v]_B)_i$$

5.9. Trace

Definition 5.9.1 (Trace of a matrix)

The trace of an $n \times n$ matrix A is the sum of the diagonal elements of A:

$$\operatorname{tr}(A) := \sum_{i=1}^{n} A_{ii}$$

$\operatorname{tr} : \mathbb{F}^{n,n} \to \mathbb{F}$ is a linear functional.

Theorem 5.9.2 (Properties of the trace)

Let A, B be $n \times n$ matrices and c be a scalar. Then
- $\operatorname{tr}(A + B) = \operatorname{tr}(A) + \operatorname{tr}(B)$
- $\operatorname{tr}(cA) = c \operatorname{tr}(A)$
- $\operatorname{tr}(A^T) = \operatorname{tr}(A)$
- $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ (this generalizes to more than 2 factors)

- The trace is similarity-invariant: if A and B are similar then $\operatorname{tr}(A) = \operatorname{tr}(B)$

5.10. Transpose

Definition 5.10.1 (Transpose of linear map)

Let V and W be vector spaces over $\mathbb{F}$, and let $T \in \mathcal{L}(V, W)$. The **transpose of T** or **dual map of T** is the linear map $T^* = T^{\mathrm{T}} \in \mathcal{L}(W^*, V^*)$ defined for each $f \in W^*$ by

$$T^*(f) = T^{\mathrm{T}}(f) \coloneqq f \circ T$$

That is, for each linear functional $f \in W^*$, $T^{\mathrm{T}}(f)$ is the linear functional in V^* defined by (for each $v \in V$)

$$T^{\mathrm{T}}(f)(v) = f(Tv)$$

Theorem 5.10.2 (Properties of the transpose)

For all linear maps T_1, T_2, T and $\alpha \in \mathbb{F}$,
- $(T_1 + T_2)^T = \alpha T_1^T + T_2^T$ (the map $T \mapsto T^T$ is linear)
- $(T_1 \circ T_2)^T = T_2^T + T_1^T$

Theorem 5.10.3 (Matrix of transpose is transpose of matrix)

The matrix of T^T is the transpose of the matrix of T:

$$[T^T]_{B_W^* B_V^*} = \left([T]_{B_V B_W}\right)^T$$

Lemma 5.10.4

Let V and W be finite-dimensional vector spaces, $T \in \mathcal{L}(V, W)$ and $g \in W^*$. Choose ordered bases $B_V = (v_1, ..., v_n)$ and $B_W = (w_1, ..., w_m)$ for V and W, respectively, and let $B_V^* = (f_1, ..., f_n)$ and $B_W^* = (g_1, ..., g_m)$ be the corresponding dual bases of V^* and W^*. Let $S = (1)$ be the standard ordered basis of $\mathbb{F}$. Then

$$[T^T g]_{B_V^*, S} = [g]_{B_W^*, S} [T]_{V,W}$$

Lemma 5.10.5

Let V and W be finite-dimensional vector spaces, $T \in \mathcal{L}(V, W)$, $T^T \in \mathcal{L}(W^*, V^*)$. Then:
- $\operatorname{rank} T^T = \operatorname{rank} T \leq \min\{\dim V, \dim W\}$
- T is injective iff T^T is surjective
- T is surjective iff T^T is injective

6. Multilinear algebra and determinants

6.1. Bilinear forms

Definition 6.1.1 (Bilinear form)

Let V be a vector space over $\mathbb{F}$. A **bilinear form** on V is a function $B : V \times V \to \mathbb{F}$ which is linear in each variable separately when the other variable is held constant. That is, for all $v_1, v_2, v \in V$ and $\alpha B(v, v_1) + B(v, v_2)$,

$$B(\alpha v_1 + v_2, v) = \alpha B(v_1, v) + B(v_2, v)$$

and

$$B(v, \alpha v_1 + v_2) = \alpha B(v, v_1) + B(v, v_2)$$

Lemma 6.1.2

If V is a vector space over $\mathbb{F}$ and $f, g \in \mathcal{L}(V, \mathbb{F})$, then $B(u, v) \coloneqq f(u)g(v)$ is a bilinear form on V.

Definition 6.1.3

$V^{(2)}$ denotes the set of all bilinear forms on V.

Lemma 6.1.4

$V^{(2)}$ is a subspace of $V \times V \to \mathbb{F}$.

Definition 6.1.5 (Matrix of a bilinear form)

Let B be a bilinear form on V and let $\vec{e} = (e_1, ...e_n)$ be an ordered basis of V. Then the matrix of B with respect to $\vec{e}$ is the matrix $[B]$ defined by:

$$[B]_{ij} = B(e_i, e_j)$$

Theorem 6.1.6

If B is a bilinear form on V, $[B]$ is its matrix with respect to the ordered basis $\vec{e}$, $v, w \in \mathbb{F}$, and $[v]$ and $[w]$ are the coordinate vectors of v and w with respect to the ordered basis $\vec{e}$, then

$$B(v, w) = [v]^T [B] [w]$$

6.2. Symmetric bilinear forms

Definition 6.2.1 (Symmetric bilinear form)

A bilinear form $B \in V^{(2)}$ is symmetric iff $B(u, w) = B(w, u)$ for all $u, w \in V$. The set of all symmetric bilinear forms on V is denoted by $V^{(2)}_{\mathrm{sym}}$.

Definition 6.2.2 (Symmetric matrix)

An $n \times n$ matrix A is symmetric iff $A^T = A$.

Theorem 6.2.3

If $B \in V^{(2)}$, then the following conditions are equivalent.
- B is a symmetric bilinear form on V.
- $[B]$ is a symmetric matrix for every basis of V.
- $[B]$ is a symmetric matrix for some basis of V.
- $[B]$ is a diagonal matrix for some basis of V.

6.3. Alternating bilinear forms

Definition 6.3.1 (Alternating bilinear form)

A bilinear form $B \in V^{(2)}$ is alternating if

$$B(u, w) = -B(w, u)$$

for all $u, w \in V$. The set of alternating bilinear forms on V is denoted by $V^{(2)}_{\text{alt}}$.

Definition 6.3.2 (Antisymmetric matrix)

A matrix A is antisymmetric iff $A^T = -A$.

Theorem 6.3.3

If $B \in V^{(2)}$, then the following conditions are equivalent.

- B is an alternating bilinear form on V.
- $[B]$ is an antisymmetric matrix for every basis of V.
- $B(v, v) = 0$ for all $v \in V$

Lemma 6.3.4

$V^{(2)} = V^{(2)}_{\text{sym}} \oplus V^{(2)}_{\text{alt}}$, and $V^{(2)}_{\text{sym}}$ and $V^{(2)}_{\text{alt}}$ are subspaces of $V^{(2)}$.

6.4. Multilinear forms

Definition 6.4.1

$$V^m := \underbrace{V \times V \times \cdots \times V}_{m \text{ times}}$$

Definition 6.4.2 (Multilinear form)

An m-linear form on V is a map which is linear in each entry when all other entries are held fixed.

The set of m-linear forms on V is denoted by $V^{(m)}$.

6.5. Alternating multilinear forms

Definition 6.5.1 (Alternating multilinear form)

An m-linear form M is alternating iff

$$M(v_1, ..., v_i, ..., v_j, ...v_m) = -M(v_1, ..., v_j, ..., v_i, ...v_m)$$

for all i, j.

Lemma 6.5.2

$M \in V^{(m)}$ is alternating iff $v_i = v_j$ implies $M(v_1, ..., v_m) = 0$ for all $i \neq j \in [1, m]$.

Lemma 6.5.3

If $M \in V^{(m)}_{\text{alt}}$ and $\{v_1, ..., v_m\}$ is linearly dependent, then $M(v_1, ..., v_m) = 0$.

Lemma 6.5.4

If $m > \dim V$, then $\dim V^{(m)}_{\text{alt}} = 0$.

6.6. Permutations

Definition 6.6.1 (Permutation)

A permutation of the set $A = \{1, 2, ..., m\}$ is a bijection $\sigma : A \to A$ (equivalently, a reordering of the ordered list A). It is often denoted by a matrix

$$\begin{bmatrix} 1 & 2 & \cdots & m \\ \sigma(1) & \sigma(2) & \cdots & \sigma(m) \end{bmatrix}$$

where the inputs form the first row and the corresponding outputs form the second row.

Definition 6.6.2 (Group)

A group is a set G which contains a binary operation $\cdot$ such that

- **Identity**: There exists an identity element $e \in G$ such that $a \cdot e = a$
- **Inverse**: There exists an inverse a^{-1} for each $a \in G$ such that $a \cdot a^{-1} = I$
- **Associativity**: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$

Theorem 6.6.3

The set S_m of all permutations of $A = \{1, 2, ..., m\}$ is a group under composition. The identity element is the trivial permutation $x \mapsto x$:

$$\begin{bmatrix} 1 & 2 & \cdots & m \\ 1 & 2 & \cdots & m \end{bmatrix}$$

The inverse of a permutation is denoted σ^{-1}, and the composition of two permutations is denoted $\tau \circ \sigma = \tau\sigma$.

Definition 6.6.4 (Cyclic permutation, r-cycle)

A **cyclic permutation** or **r-cycle** is a permutation defined by and denoted by a list $(a_1\ a_2 \cdots a_r)$ where the a_i are distinct, such that $\sigma(a_i) = a_{i+1}$ for $i \in [1, r]$, $\sigma(a_r) = a_1$, and $\sigma(a_l) = a_l$ for $l > r$.

A 2-cycle is also called a **transposition**.

Definition 6.6.5 (Disjoint cycles)

Cycles which share no elements in common.

Theorem 6.6.6

Any permutation σ can be written as a composition of disjoint transpositions.

Definition 6.6.7 (Even, odd, sign)

A permutation σ is **even** and $\text{sgn}(\sigma) = +1$ iff it is a product of an even number of disjoint transpositions.

It is **odd** and $\text{sgn}(\sigma) = -1$ iff it is a product of an odd number of disjoint transpositions.

Procedure 6.6.8 (Writing a permutation as a series of disjoint cycles)

Take any element a of the input set A, and apply σ repeatedly until $a = \sigma\cdots\sigma(a)$. Then we have cycle $a \mapsto \sigma(a) \mapsto \cdots \mapsto \sigma\cdots\sigma(a) = a$. Repeat for all remaining elements that are not part of this cycle.

Procedure 6.6.9 (Writing a permutation as a series of disjoint transpositions)

Apply the above procedure to the permutation. Then for each of the disjoint cycles in the series that are not transpositions, apply that procedure to each of those cycles until they are transpositions (2-cycles).

Lemma 6.6.10

$\operatorname{sgn}(\tau\sigma) = \operatorname{sgn}(\tau)\operatorname{sgn}(\sigma)$ for all $\tau, \sigma \in S_m$.

Lemma 6.6.11

An r-cycle is an even permutation iff r is odd, and an odd permutation iff r is even.

Procedure 6.6.12 (Determining the sign of a permutation)

Decompose the permutation into a product of disjoint cycles. Then the parity (oddness/evenness) of the permutation is the number of cycles of even length in its decomposition.

6.7. Determinant

Definition 6.7.1

$[v_1 \cdots v_n]$ is the matrix whose jth column is v_j.

$|v_1 \cdots v_n|$ is the determinant of said matrix.

Definition 6.7.2 (Determinant of a linear operator)

For $T \in \mathcal{L}(V)$, the **determinant** of T is the unique scalar such that $M_T := M \circ T = (\det T)M$ for all $M \in V_{\text{alt}}^{(\dim V)}$.

Definition 6.7.3 (Determinant of a square matrix)

Let n be a positive integer, A be an $n \times n$ matrix with entries in $\mathbb{F}$, and $T \in \mathcal{L}(\mathbb{F}^n)$ be the operator whose matrix with respect to the standard basis of $\mathbb{F}^n$ is A. Then the determinant of A is $\det A := \det T$.

Lemma 6.7.4

Let $(v_1, \ldots, v_n)$ be an ordered n-tuple of column vectors. Then the map $(v_1, \ldots, v_n) \mapsto |v_1 \cdots v_n|$ is an alternating n-linear form on $\mathbb{F}^n$.

Lemma 6.7.5 ($O(n!)$ method for calculating determinant)

If A is an $n \times n$ matrix,

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) A_{\sigma(1)1} \cdots A_{\sigma(n)n}$$

Theorem 6.7.6 (Cofactor method (also $O(n!)$))

Let A be an $n \times n$ matrix. Define $A[i|j]$ to be the result of removing the ith row and jth column from A. Define the ***ij*-cofactor** of A to be

$$C_{ij} := (-1)^{i+j} \det(A[i|j])$$

Then for any row numbered i,

$$\det A = \sum_{k=1}^{n} A_{ik} C_{ik}$$

And for any column numbered j,

$$\det A = \sum_{k=1}^{n} A_{kj} C_{kj}$$

Note that the coefficient $(-1)^{i+j}$ in C_{ij} forms the following pattern:

$$\begin{bmatrix} + & - & + & \cdots \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

Lemma 6.7.7

If A is a triangular matrix with $\lambda_1, \ldots, \lambda_n$ on the diagonal, then $\det A = \lambda_1 \cdots \lambda_n$ ($\det A$ is the product of the elements on the diagonal).

6.8. Properties of the determinant

Theorem 6.8.1 (Determinant is multiplicative)

- If $S, T \in \mathcal{L}(V)$, then $\det(ST) = (\det S)(\det T)$.
- If A and B are $n \times n$ matrices, then $\det(AB) = (\det A)(\det B)$.

Theorem 6.8.2

A linear operator $T \in \mathcal{L}(V)$ is an isomorphism iff $\det T \neq 0$.

If T is an isomorphism, then $\det(T^{-1}) = (\det T)^{-1}$.

Theorem 6.8.3

The following are equivalent:

- A is invertible.
- $\det A \neq 0$.
- The homogeneous system $AX = 0$ has only the unique solution.

Theorem 6.8.4 (Determinant is similarity invariant)

Let $T \in \mathcal{L}(V)$ and $S : W \to V$ be an invertible linear map. Then

$$\det(S^{-1}TS) = \det T$$

Theorem 6.8.5

For all $T \in \mathcal{L}(V)$, $\det T = \det[T]$, where $[T]$ is the matrix of T with respect to any basis of V.

Theorem 6.8.6

If A is a square matrix, or if $A \in \mathcal{L}(V)$, then

$$\det(A^T) = \det(A)$$

Theorem 6.8.7 (Effect of row and column operations on the determinant)

Let A be an $n \times n$ matrix.

- If two rows or columns of A are equal, then $\det A = 0$.
- If B is the result of swapping two rows or columns of A, then $\det B = -\det A$.
- If B is the result of multiplying one row or column of A by the scalar λ, then $\det B = \lambda \det A$
- If B is the result of replacing a column or row by the sum of itself and a scalar multiple of another column or row, then $\det B = \det A$. (this does not apply adding a column to a scalar multiple of a row or vice-versa.)

Theorem 6.8.8 (Effect of elementary row operations (summary))

Let A be an $n \times n$ matrix.

- **Scaling**: $R_i \mapsto cR_i \Rightarrow \det B = c \det A$.
- **Replacement**: $R_i \mapsto R_i + cR_j \Rightarrow \det B = \det A$
- **Interchange**: $R_i \leftrightarrow R_j \Rightarrow \det B = -\det A$

Procedure 6.8.9 ($O(n^3)$ procedure for finding determinant)

Row-reduce the matrix A to row-echelon form ($R := \mathrm{REF}(A)$). Then $\det R$ is the product of the items along the diagonal of R. To find $\det A$, apply the above theorems.

6.9. Cramer's rule

Definition 6.9.1 (Adjugate matrix)

Let A be an $n \times n$ matrix. Then the adjugate matrix of A is the transpose of the matrix of cofactors, i.e.

$$\operatorname{adj} A = C^T$$

where C is defined by

$$C_{ij} = (-1)^{i+j} \det(A[i|j])$$

Lemma 6.9.2

Let A be an invertible $n \times n$ matrix. Then

$$A^{-1} = \frac{1}{\det A} \operatorname{adj} A$$

Theorem 6.9.3 (Cramer's Rule)

Let A be an invertible $n \times n$ matrix. For any $\vec{b} \in \mathbb{R}^n$, the unique solution $\vec{x}$ of the linear system $A\vec{x} = \vec{b}$ has entries given by

$$x_i = \frac{\det A_i(\vec{b})}{\det A}$$

where $A_i(\vec{b})$ is the result of replacing the ith column of A by $\vec{b}$.

7. Eigenvalues and eigenvectors

7.1. Eigenvalues and eigenvectors

Theorem 7.1.1 (Diagonality condition)

Let V be a finite-dimensional vector space, $B = (v_1, ..., v_n)$ be an ordered basis for V, and $T \in \mathcal{L}(V)$. Then $[T]_B$ is diagonal iff $T(v_i) = \lambda_i v_i$ for all i, where λ_i is the ith element along the diagonal of $[T]_B$.

Definition 7.1.2 (Eigenvalue, eigenvector of a linear operator)

Let V be a vector space and $T \in \mathcal{L}(V)$. An **eigenvalue** of T is a scalar λ s.t. there exists a nonzero vector $v \in V$ s.t.

$$T(v) = \lambda v$$

The vector v is the **eigenvector** of T associated with the eigenvalue λ.

Lemma 7.1.3

If v is an eigenvector of T associated with eigenvalue λ and if B is a basis of V, then

$$[T(v)]_B = \lambda [v]_B$$

Definition 7.1.4 (Eigenvalue, eigenvector of a square matrix)

An **eigenvalue** of the $n \times n$ matrix A is a scalar λ s.t. there exists a nonzero vector $v \in \mathbb{R}^n$ s.t.

$$Av = \lambda v$$

The vector v is the **eigenvector** of A associated with the eigenvalue λ.

Lemma 7.1.5

Similar matrices have the same eigenvalues.

Any matrix representing $T \in \mathcal{L}(V)$ has the same eigenvalues as T.

Definition 7.1.6 (Eigenspace)

A scalar λ is an eigenvalue of A iff $Av = \lambda v$ has a nontrivial solution. This is equivalent to the homogeneous linear system (where I is the $n \times n$ identity matrix)

$$(A - \lambda I)v = 0$$

The set of all solutions to this equation is the **eigenspace** of A associated to the eigenvalue λ, which we denote by E_λ.

Lemma 7.1.7

If $A \in \mathbb{R}^{n\times n}$, then the eigenspace of A corresponding to λ is a subspace of $\mathbb{R}^n$.

Lemma 7.1.8

Let A be an $n \times n$ matrix. Then $\det(\lambda I - A)$ is a monic (leading coefficient 1) polynomial of degree n on λ.

Definition 7.1.9

For any $n \times n$ matrix A, the monic, degree n polynomial $\det(\lambda I - A)$ is the **characteristic polynomial** of A. The equation

$$\det(\lambda I - A) = 0$$

is the **characteristic equation** of A.

Lemma 7.1.10

Similar matrices have the same characteristic equation.

Lemma 7.1.11

The eigenvalues of a triangular matrix are the entries on the main diagonal.

Theorem 7.1.12 (Rational root theorem)

Let $p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ be a degree-n polynomial with integer coefficients. If $x = \frac{p}{q}$ is a rational root of $p(x)$, then a_0 is divisible by p and a_n is divisible by q.

Procedure 7.1.13 (Computing eigenvalues)

Firstly, if the matrix is triangular, then the eigenvalues are the entries on the main diagonal.

Otherwise, solve the characteristic equation for λ to get the eigenvalues:

$$\det(\lambda I - A) = 0$$

Plug each eigenvalue into the equation

$$(A - \lambda I)v = 0$$

and the solution set is the eigenspace. Pick any vector in the eigenspace to be your eigenvector.

7.2. Diagonalizable operators

Definition 7.2.1 (Diagonalizable operator)

A linear operator $T \in \mathcal{L}(V)$ is said to be diagonalizable if there exists a basis consisting of eigenvectors of T. Such a basis is called an **eigenbasis**.

Lemma 7.2.2

If $T \in \mathcal{L}(V)$, then every set of eigenvectors corresponding to distinct eigenvalues of T is linearly independent.

Lemma 7.2.3

If $\dim V = n$ and $T \in \mathcal{L}(V)$ has n distinct eigenvalues, then T is diagonalizable.

Definition 7.2.4 (Multiplicity of a root of a polynomial)

Let $p(x)$ be a polynomial. Then an element $a \in \mathbb{F}$ is a root of multiplicity k iff there exists a polynomial $s(x)$ such that $s(a) \neq 0$ and $p(x) = (x-a)^k s(x)$.

Definition 7.2.5 (Algebraic multiplicity)

The algebraic multiplicity of the eigenvalue λ_k is the multiplicity of λ_k as a root of the characteristic polynomial of A.

Definition 7.2.6 (Geometric multiplicity)

The geometric multiplicity of the eigenvalue λ_k is the dimension of the eigenspace corresponding to λ_k.

Theorem 7.2.7 (Conditions equivalent to diagonalizability)

Let V be a finite-dimensional vector space, $T \in \mathcal{L}(V)$, and $\lambda_1, ..., \lambda_n$ be the distinct eigenvalues of T. Then the following are equivalent:

- V has a basis consisting of eigenvectors of T
- $V = E_{\lambda_1} \oplus \cdots \oplus E_{\lambda_m}$
- $\dim V = \dim E_{\lambda_1} + \cdots + \dim E_{\lambda_m}$
- The geometric multiplicity of each eigenvector is equal to its algebraic multiplicity.

Procedure 7.2.8 (Diagonalizing a linear operator)

Let $T \in \mathcal{L}(V)$. Choose an ordering $B = (v_1, v_2, ..., v_n)$ of the eigenvectors of T, corresponding to the eigenvalues $\lambda_1, \lambda_2, ..., \lambda_n$. Then $[T]_B$, the diagonalization of T, the matrix of T with respect to B, is

$$\begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}$$

7.3. Powers of a diagonalizable operator

Lemma 7.3.1

If v is an eigenvector of T with eigenvalue λ and n is a positive integer, then $T^n v = \lambda^n v$.

Lemma 7.3.2

If T is diagonalizable, then there exists a basis D such that the diagonalization of T is equal to

$$[T]_D = C_{e\to D}[T]_e C_{D\to e}$$

(where $C_{e\to D}$ is the transition matrix from the standard basis to D)

Furthermore, for any positive integer n,

$$[T]_e^n = C_{D\to e}[T]_D^n C_{e\to D}$$

Procedure 7.3.3 (Power of a diagonal matrix)

Let D be a diagonal matrix equal to

$$\begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}$$

Then D^n is equal to

$$\begin{bmatrix} \lambda_1^n & 0 & \cdots & 0 \\ 0 & \lambda_2^n & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n^n \end{bmatrix}$$

8. Inner product spaces

For any discussion of inner product spaces, let $\mathbb{F}$ be $\mathbb{R}$ or $\mathbb{C}$.

8.1. Inner product

See Axler [6A], Trimm [7.1], MATH 257 [Module 15]

Definition 8.1.1 (Positive definite)

A map $f : V \times V \to \mathbb{F}$ is positive definite iff for any $v \in V$ where $v \neq 0$,

$$f(v, v) > 0$$

and $f(0, 0) = 0$.

Definition 8.1.2 (Inner product on a real vector space)

Let V be a real vector space. Then an inner product on V is a positive definite symmetric bilinear form on V.

The inner product of $v, w \in V$ is denoted (v, w).

Definition 8.1.3 (Norm)

If $u, v \mapsto (u, v)$ is an inner product, then the ***norm*** of a vector v is defined as

$$\|v\| \coloneqq (v, v).$$

Definition 8.1.4 (Dot product on $\mathbb{R}^n$)

The dot product is the mapping that maps $v, w \in \mathbb{R}^n$ to the real number

$$v \cdot w = \sum_{i=1}^{n} v_i w_i = v^{\mathrm{T}} w$$

It is an inner product on $\mathbb{R}^n$, and satisfies the following properties for any $v, w, x \in V$:

- $(cv + w) \cdot x = cv \cdot x + w \cdot y$
- $v \cdot w = w \cdot v$
- $x \cdot (cv + w) = cx \cdot v + x \cdot w$
- $\|v\| = \sqrt{v \cdot v}$

Definition 8.1.5 (Complex conjugate)

The complex conjugate of the complex number $a + bi$ is

$$\overline{a + bi} = a - bi$$

Definition 8.1.6 (Conjugate-linear map)

A map $f : V \to W$ where V and W are complex vector spaces is conjugate-linear iff for all $x, y \in V$, $s \in \mathbb{C}$,

- **Additivity**: $f(x + y) = f(x) + f(y)$
- **Conjugate homogeneity**: $f(sx) = \overline{s} f(x)$

A conjugate-linear map is a type of **semilinear map**.

Definition 8.1.7 (Sesquilinear map)

A map $f : V \times V \to W$ which is linear in the first argument and semilinear in the second argument.

Iff it is conjugate-linear in the second argument, for all $z_1, z_2 \in \mathbb{C}^n$, $c \in \mathbb{C}$,

$$f(w, cz_1 + z_2) = \overline{c} f(w, z_1) + f(2, z_2)$$

Definition 8.1.8 (Inner product on a complex vector space)

Let V be a complex vector space. Then an inner product on V is a positive-definite sesquilinear form on V.

Definition 8.1.9 (Dot product on $\mathbb{C}^n$)

The dot product is the mapping that maps $z, w \in \mathbb{C}^n$ to the complex number

$$z \cdot w = \sum_{i=1}^{n} z_i \overline{w_i}$$

It is an inner product on $\mathbb{C}^n$, and satisfies the following properties for any $v, w, x \in V$:

- $(cv + w) \cdot x = cv \cdot x + w \cdot y$
- $v \cdot w = \overline{w \cdot v}$
- $x \cdot (cv + w) = \overline{c}x \cdot v + x \cdot w$
- $\|v\| = \sqrt{v \cdot v}$

Definition 8.1.10 (Inner product space)

A vector space V together with an inner product on V.

Theorem 8.1.11

For all $u, v \in V$,

- **Cauchy-Schwarz inequality**: $|(u, v)| \leq \|u\|\|v\|$
- **Triangle Inequality**: $\|u + v\| \leq \|u\| + \|v\|$
- **Reverse Triangle Inequality**: $||u| - |v|| \leq \|u - v\|$
- **Parallelogram Law**: $\|u + v\|^2 + \|u - v\|^2 = 2(\|u\|^2 + \|v\|^2)$

Let V be an inner product space.

Definition 8.1.12 (Orthogonality)

Two vectors $u, v \in V$ are orthogonal iff $(u, v) = 0$.

Lemma 8.1.13

0 is orthogonal to all vectors.

Theorem 8.1.14 (Pythagorean Theorem)

If $u, v \in V$ are orthogonal, then

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2$$

Theorem 8.1.15 (Orthogonal decomposition)

Let $u, v \in V$. Then u can be uniquely decomposed as

$$u = u_{\|} + u_{\perp}$$

where $u_{\|} \parallel v$ (i.e. $u_{\|} = cv$ for some $c \in \mathbb{F}$) and $u_{\perp} \perp v$ (i.e. $(u_{\perp}, v) = 0$).

We can construct them as:

$$u_{\|} := \frac{(u, v)}{\|v\|^2} v \quad u_{\perp} := u - u_{\|}$$

Definition 8.1.16 (Unit vector)

$v \in V$ is a ***unit vector*** iff $\|v\| = 1$.

Definition 8.1.17 (Orthogonal set)

A set A where every $v, w \in A$ are orthogonal.

Definition 8.1.18 (Orthonormal set)

An orthogonal set A where every $v \in A$ has $\|v\| = 1$.

9. Orthogonal complement

See MATH 257 [Module 23]

Definition 9.1 (Orthogonal complement)

Let W be a subspace of $\mathbb{R}^n$. The ***orthogonal complement*** of W is the subspace $W^{\perp}$ of all vectors that are orthogonal to W; that is

$$W^{\perp} := \{v \in \mathbb{R}^n : v \cdot w = 0 \text{ for all } w \in W\}.$$

Remark

Observe that $\left(W^{\perp}\right)^{\perp} = W$.

Lemma 9.2 (Transpose is the adjoint of the dot product)

For all $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$,

$$(Ax) \cdot y = x \cdot (A^{\mathrm{T}} y).$$

Lemma 5.5.9

$$\mathrm{Nul}(A^{\mathrm{T}}) = \mathrm{Col}(A)^{\perp}$$
$$\mathrm{Row}(A) = \mathrm{Col}(A^{\mathrm{T}}) = \mathrm{Nul}(A)^{\perp}$$

Theorem 9.3

Let V be a subspace of $\mathbb{R}^n$. Then $\dim V + \dim V^{\perp} = n$.

10. Appendix: Graphs

Definition 10.1

A graph is a set of nodes (or: vertices) that are connected through edges.

Definition 10.2

Let G be a graph with n nodes. The adjacency matrix of G is the $n \times n$-matrix $A = \left(a_{ij}\right)$ such that

$$a_{ij} = \begin{cases} 1 \text{ if there is an edge between node } i \text{ and node } j \\ 0 \text{ otherwise} \end{cases}$$

Definition 10.3

A walk of length k on a graph is a sequence of $k + 1$ vertices and k edges between two nodes (including the start and end) that may repeat. A path is a walk in which all vertices are distinct.

Definition 10.4

A graph is connected if for every pair of nodes i and j there is a walk from node i to node j. A graph is disconnected if it is not connected.

Theorem 10.5

Let G be a graph and let A be its adjacency matrix. Then the entry in the i-th row and j-th column of A^{ℓ} is the number of walks of length ℓ from node j to node i on G.

Definition 10.6

A directed graph is a set of vertices connected by edges, where the edges have a direction associated with them.

Definition 10.7

Let G be a directed graph with m edges and n nodes. The adjacency matrix of G is the $n \times n$ matrix $A = \left(a_{i,j}\right)_{i,j}$ with

$$a_{i,j} = \begin{cases} 1 \text{ if there is a directed edge from node } j \text{ to node } i \\ 0 \text{ otherwise} \end{cases}$$

Definition 10.8

Let G be a directed graph with m edges and n nodes. The edge-node incidence matrix of G is the $m \times n$ matrix $A = \left(a_{i,j}\right)_{i,j}$ with

$$a_{i,j} = \begin{cases} -1 \text{ if edge } i \text{ leaves node } j \\ +1 \text{ if edge } i \text{ enters node } j \\ 0 \quad \text{otherwise} \end{cases}$$

Definition 10.9

A connected component of an undirected graph is a part in which any two vertices are connected to each other by paths, and which is connected to no additional vertices in the rest of the graph. The connected components of a directed graph are those of its underlying undirected graph. A graph is connected if it has only one connected component.

Theorem 10.10

Let G be a directed graph and let A be its edge-node incidence matrix. Then $\dim \mathrm{Nul}(A)$ is equal to the number of connected components of G.

Definition 10.11

A cycle in an undirected graph is a path in which all edges are distinct and the only repeated vertices are the first and last vertices. By cycles of a directed graph we mean those of its underlying undirected graph.

Definition 10.12

The span of all cycle vectors of a graph G is called the cycle space.

Theorem 10.13

Let G be a directed graph and let A be its edge-node incidence matrix. Then the cycle space of G is equal to $\mathrm{Nul}(A^T)$.

Why the walk-counting theorem is true (case $\ell = 2$)

Write A's rows as $R_1, \ldots, R_n$ and columns as $C_1, \ldots, C_n$. By the row-column rule,

$$\left(A^2\right)_{ij} = R_i \cdot C_j = A_{i1}A_{1j} + A_{i2}A_{2j} + \cdots + A_{in}A_{nj}.$$

Each summand $A_{ik}A_{kj}$ equals 1 exactly when there is an edge from j to k **and** an edge from k to i — i.e. exactly when it witnesses a walk of length 2 from j to i passing through k. Since every walk of length 2 passes through exactly one intermediate node, the sum counts all such walks. The same argument, with $A^{\ell-1}$ in place of A, extends this to any ℓ by induction.

Lemma 10.14

The adjacency matrix of an undirected graph is symmetric: $A^{\mathrm{T}} = A$. A nonzero diagonal entry A_{ii} indicates a ***self-loop*** at node i.

Remark

Since $(I)_{ij} = 1$ exactly when $i = j$, it is natural to regard I as counting the (trivial) walk of length 0 from a node to itself. Consequently $I + A$ counts walks of length 0 or 1 between each pair of nodes — and more generally, $I + A + A^2 + \cdots + A^k$ counts walks of length at most k.

Theorem 10.15 (Walks of length at most k)

Let G be a graph with adjacency matrix A. Every node of G is connected to every other node by a walk of length at most k if and only if

$$I + \sum_{i=1}^{k} A^i$$

has no zero entries.

Remark

Two subtleties:

- Using $\sum_{i=1}^{k} A^i$ **without** the I term counts only nontrivial walks between each pair; it can still have zero diagonal entries even for a connected graph, since a node need not have a walk back to itself of length $\leq k$.
- Using A^k **alone** does not correctly count "reachable within k steps": a node reached in fewer than k steps is forced to leave again at the next step, unless A already has a self-loop at every node (cf. $A + I$).

Corollary 10.15.1

A graph on n nodes is connected if and only if $I + \sum_{i=1}^{n-1} A^i$ has no zero entries — since any walk connecting two distinct nodes without revisiting a node uses at most $n - 1$ edges, taking $k = n - 1$ always suffices to test full connectivity.

Potential interpretation of $\mathrm{Nul}(M)$

Let M be the edge-node incidence matrix of a directed graph G with n nodes, and think of $\vec{x} \in$

$\mathbb{R}^n$ as assigning a ***potential*** x_k to each node k. For the row of M corresponding to an edge e from node j to node i,

$$(M\vec{x})_e = x_i - x_j,$$

so $M\vec{x} = 0$ forces the potentials at the two endpoints of every edge to agree. Hence $\vec{x} \in \mathrm{Nul}(M)$ assigns equal potential to every pair of nodes joined by a path — i.e. constant potential on each connected component. This gives another route to $\dim \mathrm{Nul}(M) =$ number of connected components: a basis is given by the indicator vector of each connected component.

Flow interpretation of $\mathrm{Nul}(M^{\mathrm{T}})$

Think of $\vec{y} \in \mathbb{R}^m$ (m = number of edges) as assigning a flow y_e to each edge. For node k,

$$(M^{\mathrm{T}}\vec{y})_k = (\text{total inflow at } k) - (\text{total outflow at } k),$$

so $\vec{y} \in \mathrm{Nul}(M^{\mathrm{T}})$ exactly when inflow equals outflow at every node — ***Kirchhoff's current law***. This is why the cycle space (flows that balance at every node) equals $\mathrm{Nul}(M^{\mathrm{T}})$.

Procedure 10.16 (Writing a cycle as a vector)

Given a cycle in G and a chosen direction to traverse it, build $\vec{c} \in \mathbb{R}^m$ by setting, for each edge e:

- $c_e = +1$ if the cycle traverses e in e's own defined direction (tail to head),
- $c_e = -1$ if traversed against it,
- $c_e = 0$ if e is not part of the cycle.

Any such $\vec{c}$ automatically satisfies $M^{\mathrm{T}}\vec{c} = 0$ (flow is conserved at every node along the cycle).

Remark

Cycle vectors built this way need not be independent — e.g. for three cycles that share edges, one may find $\vec{c}_1 + \vec{c}_2 = \vec{c}_3$. As with any null space, a genuine **basis** for the cycle space is found by solving $M^{\mathrm{T}}\vec{y} = 0$ with the usual homogeneous-system procedure (see [Linear systems and matrices]), not by listing cycles directly.

11. Resources

11.1. Main resources

- **Trimm** – notes from "TrimmAlg", the linear algebra class taught by Dr. Anderson Trimm in S25
  - **Karthik** – a student's solutions to worksheets from TrimmAlg
  - This class follows Axler, but presents some things from Strang in an Axler-style presentation
- **MATH 257** – notes from MATH 257 "Linear Algebra with Computational Applications", taught in F26
  - **Chaung** – notes from the 09:00 lecture given by Jer-Chin Chuang in F26
  - This is a more numerical/applied class
  - This class mostly follows Strang and MIT 18.06, but also introduces worked examples in NumPy code
- **Axler** – *Linear Algebra Done Right*, 4th edition, by Sheldon Axler
  - Solution manual by GitHub user `@lew98`
  - A very abstract book

11.2. Supplementary resources

- **Klein** – *Coding the Matrix: Linear Algebra through Applications to Computer Science*, by Philip N. Klein
  - Philip Klein's lecture videos
  - Eng Weam's lecture videos
  - Coursera videos
- **UCDavis** – open-access textbook *Linear Algebra*, by Cherney, Denton, Thomas, and Waldron
  - Strang-like approach
- **Alayont & Schlicker** – Linear Algebra and Applications: An Inquiry-Based Approach
  - Inquiry-based approach, i.e. investigation of examples

11.2.1. Gilbert Strang's main curriculum

- **Strang-LAIA** – *Linear Algebra and Its Applications*
  - Strang's oldest textbook
- **Strang-ILA** – *Introduction to Linear Algebra*
  - The book in Strang's course MIT 18.06
  - Website of Fourth Edition (2009)
  - Website of Fifth Edition (2016)
  - Website of Sixth Edition (2023)
    - It has useful links and some full chapters
- **Strang-LALD** – *Linear Algebra and Learning from Data*
  - Website of First Edition (2019)
- **MIT 18.06**
  - Main website
  - OpenCourseWare (Spring 2010)
  - **MIT 18.06SC** (OpenCourseWare)
    - Designed for self-study
    - Has lecture notes on OCW, unlike 18.06
    - Has problem-solving help videos
  - *ZoomNotes for Linear Algebra*
  - 18.06 Related Resources
- **MIT 18.065** "Matrix Methods in Data Analysis, Signal Processing, and Machine Learning"
  - MIT 18.065, Spring 2018
  - MIT 18.065, Spring 2023

11.2.2. Gilbert Strang's new curriculum (CR-factorization)

- **Strang-LAFE** – *Linear Algebra for Everyone*
  - Website of First Edition (2020)
  - *The Art of Linear Algebra* by Kenji Hiranabe – graphical representations of stuff from LAFE

- *A Vision of Linear Algebra* videos (2020~2024)

11.3. Videos

11.3.1. 3Blue1Brown

1. Vectors
   - $\mathbb{R}^2$ and $\mathbb{R}^3$ (geometrically)
   - Vectors as arrays and as arrows from the origin
   - Vector addition in $\mathbb{R}^2$ and $\mathbb{R}^3$
2. Linear combinations, span, and basis vectors
   - Basis (Section 2.7)
   - Spanning (Section 2.5)
   - Subspaces of $\mathbb{R}^2/\mathbb{R}^3$ (Section 2.3)
3. Linear transformations and matrices
   - Linear maps (Section 3.1)
   - Matrix of a linear map (Section 4.3)
   - Matrix-vector multiplication (Section 4.5)
4. Matrix multiplication as composition
   - Matrix multiplication (Section 4.10)
5. Three-dimensional linear transformations
6. The determinant
   - TODO
7. Inverse matrices, column space and null space
   - Linear systems (Section 4.4)
   - Invertible matrices (Section 5.2)
   - Column and null space (Section 5.5)
8. Nonsquare matrices as transformations between dimensions
9. Dot products and duality
   - Inner product (Section 8.1)
   - TODO: geometric meaning of the dot product, relationship between row vector as transformation and vector as a vector